ANUSHKA KATIYAR

Los Angeles, USA | 213-284-1984 | akatiyar@usc.edu | LinkedIn | Github

EDUCATION

University of Southern California **Los Angeles, CA**
Master of Science, Electrical & Computer Engineering – Machine Learning & Data Science (GPA 3.8/4.0) Jan 2025 – Dec 2026

Harcourt Butler Technical University **Kanpur, India**
Bachelor of Technology, Electronics Engineering, Graduated with Honors June 2018 – May 2022

WORK EXPERIENCE

National Laboratory of the Rockies (prev. NREL) **Golden, CO**
Machine Learning Intern May 2026 – Present

- Developed a physics-informed probabilistic solar forecasting framework using **Mixture-of-Experts (MoE)**, clear-sky modeling, and **deep learning** to reduce intra-hour PV forecast error by 15%+ and improve reserve market bid accuracy.
- Built a probabilistic PV forecasting pipeline integrating cloud tracking, inverter spatio-temporal data, and vendor forecast fusion, improving quantile forecast accuracy by 20%.

University of Southern California **Los Angeles, CA**
Graduate Research Assistant
Wireless Devices & Systems Lab(WiDES) (*Lab Website*) Sep 2025 – Present

- Designed **CNN & transformer** pipelines for indoor Wi-Fi localization from spectral features, improving accuracy by ~25%.
- Applied **LoRA-based parameter-efficient fine-tuning** to the localization transformer, cutting trainable parameters by 98% and achieving ~2.37m RMSE across 3 buildings and 11 devices, enabling fast adaptation to new environments without full retraining.

Energy Efficient Secure Sustainable Computing Group (E2S2C) (*Lab Website*) Sep 2025 – Feb 2026

- Developed **agentic AI** pipeline with Gemini 2.5 API for automated Verilog code generation, leveraging the **NVIDIA CVDP** Benchmark Dataset and Hugging Face frameworks achieving 47% one-shot accuracy.
- Designed and evaluated **multi-agent architectures** to enhance scalability and efficiency in hardware description language synthesis and automation reducing context window tokens by 50% through reinforcement-optimized policies.

Solace Technologies **Marina Del Ray, CA**
AI/ML Engineer, Intern (Demo Video) May 2025 – Aug 2025

- Prototyped an AI-driven cost and schedule prediction MVP for aerospace projects using **XGBoostRegressor** on engineered features, achieving 93% prediction accuracy across 200+ projects.
- Enhanced prediction interpretability and scalability by integrating **RAG with Mistral 13B, FAISS, and LangChain** for unstructured document retrieval, and deploying ML pipelines with **Airflow, MLflow, and AWS SageMaker**.

Bharti Airtel **Gurugram, India**
Data Scientist May 2024 – Oct 2024

- Deployed **LSTM** fiber monitoring model, reducing network downtime by 30% and automating performance diagnostics at scale.
- Automated customer communication workflows by standardizing templates and integrating **Google's BERT**-powered NLP model for intelligent processing of resolution insights, resulting in a 25% decrease in customer interactions.
- Leveraged sentiment analysis and topic modeling techniques (**LDA, TF-IDF**) on 80,000 monthly customer complaints to proactively identify emerging fault patterns, boosting targeted maintenance capabilities by 15%.

Data Analyst Aug 2022 – April 2024

- Architectured and deployed nationwide tower companies growth/degrowth prediction model using **ARIMA** and regression analysis, achieving 93% accuracy and driving strategic planning and resource optimization by 45%.
- Spearheaded the design and launch of an assurance analytics portal in collaboration with IBM, boosting operational productivity by 15% and enabling real-time **KPI** tracking for ~1 million B2B customers, used across 25 circle offices.

PROJECTS

REFLECT-RL:Reinforcement Learning Framework for Autonomous Code Generation (Github**)**

- Engineered a self-improving reinforcement learning framework for autonomous code synthesis using **FAISS-based RAG**, iterative feedback loops, and **CodeLlama 7B** for adaptive reasoning.
- Applied **Reflexion-Policy Optimization (RPO)** to dynamically refine agent performance, achieving a 62% improvement in **HumanEval benchmark** accuracy across 161 programming tasks, establishing a scalable pipeline for multi-model inferencing.

Weather-Driven Collision Risk Forecasting in NYC| Northeast Big Data Innovation Hub & Columbia University (Github**)**

- Performed advanced **time series and spatio-temporal analysis** on 1M+ NYC collision records as part of the National Student Data Corps, integrated **OpenWeather API** to correlate 500k+ weather data points with collision rates, identifying 20% higher crash likelihood during rain, and proposed safety improvements to the DOT.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, MATLAB
Machine Learning & AI: PyTorch, TensorFlow ,Scikit-learn, XGBoost, Hugging Face, LangChain, FAISS, OpenCV, RAG, LoRA
Data & MLOps: Pandas, NumPy, Docker, Airflow, MLflow, AWS SageMaker, Git, Linux, Tableau, ETL Pipelines, REST APIs
Core Areas: Deep Learning, LLMs, NLP, Computer Vision, Time Series & Probabilistic Forecasting, Reinforcement Learning
Certifications:Machine Learning Specialization Stanford/DeepLearning.AI, Data Analytics & Virtual Reality- NASSCOM